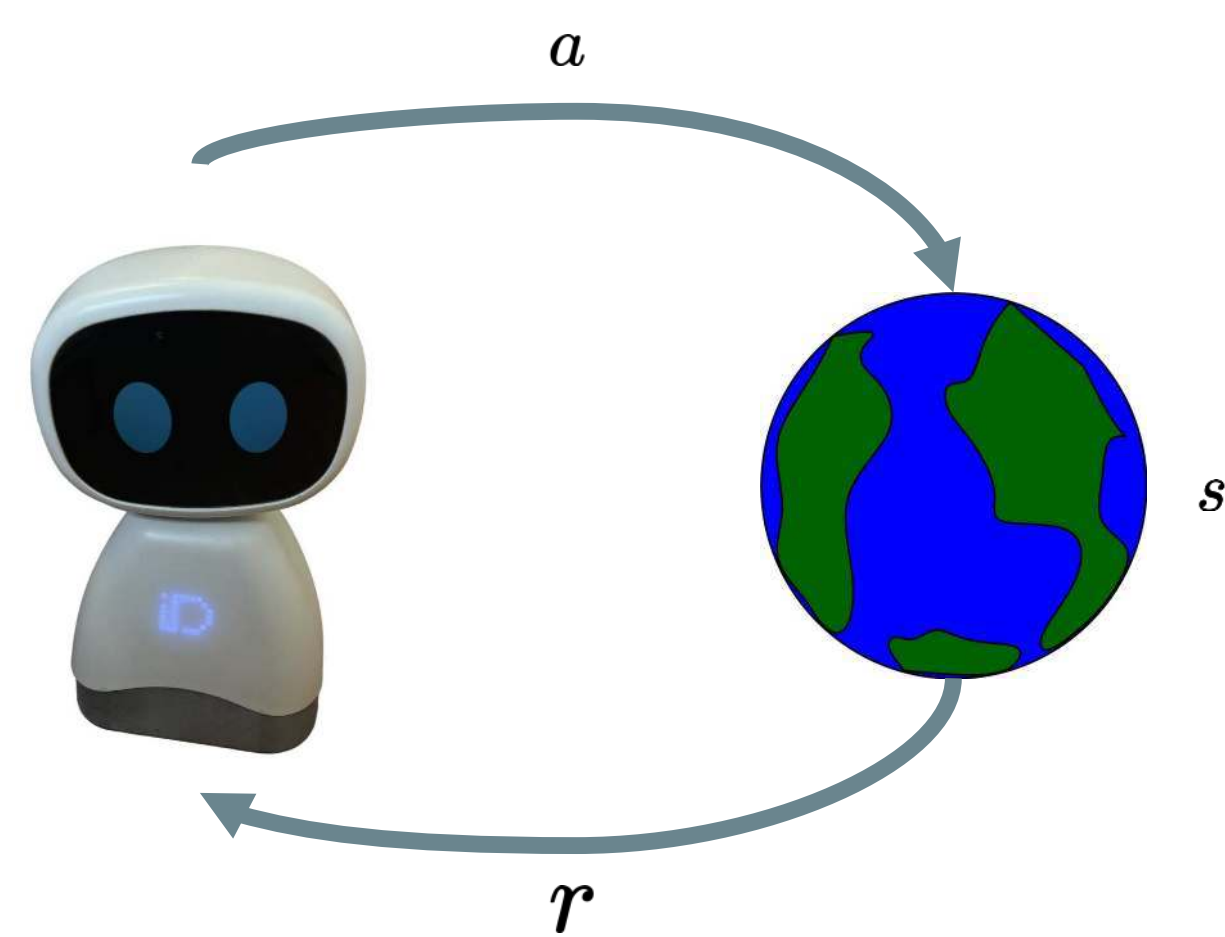


Multi-Bellman Operator for Convergence of Q-learning

Diogo S. Carvalho,* Pedro A. Santos, and Francisco S. Melo
Instituto Superior Técnico, University of Lisbon, and INESC-ID
*diogo.s.carvalho@tecnico.ulisboa.pt

1. Introduction: Reinforcement Learning



Q-learning aims to compute q^*
the fixed-point of the Bellman operator H

2. Multi-Bellman Operator

$$H^{n+1} q = H H^n q$$

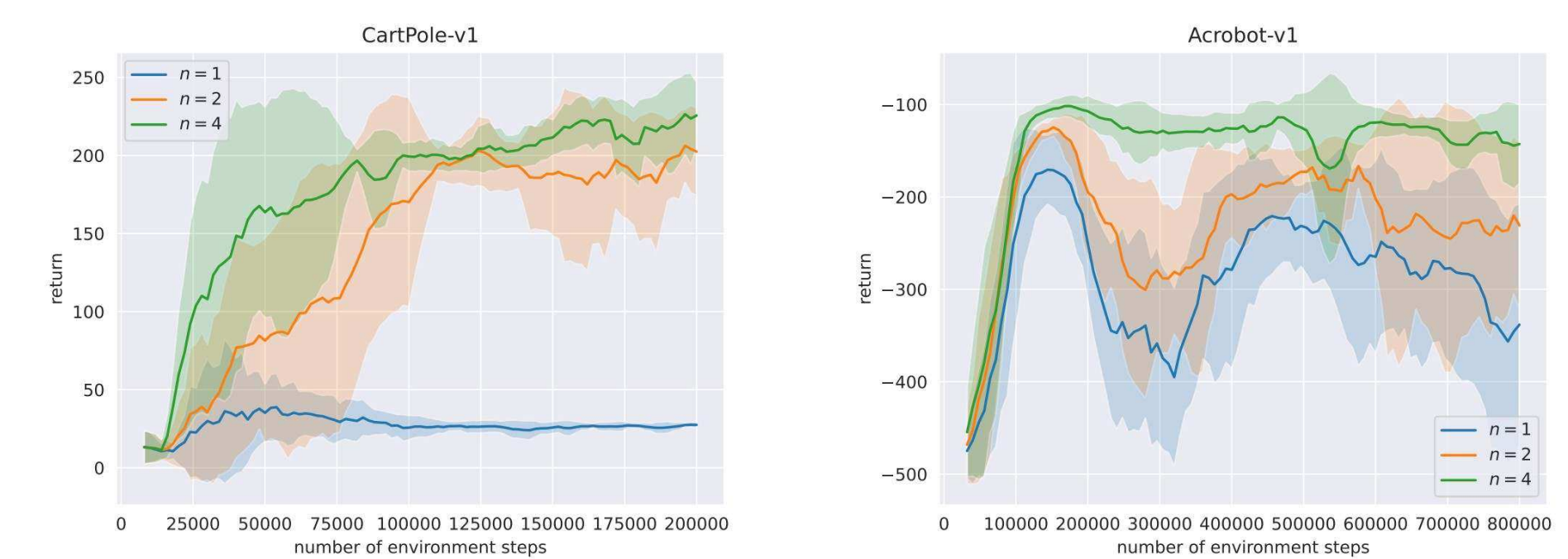
Apply the Bellman operator n
times before the projection

3. Multi Q-learning

$$\omega_{t+1} = \omega_t + \alpha_t (\tau_{n,m} - q_{\omega_t}) \nabla_{\omega} q_{\omega_t}$$

Build multi-step target with n depth and
 m breadth and then update the weights

4. Experimental Validation



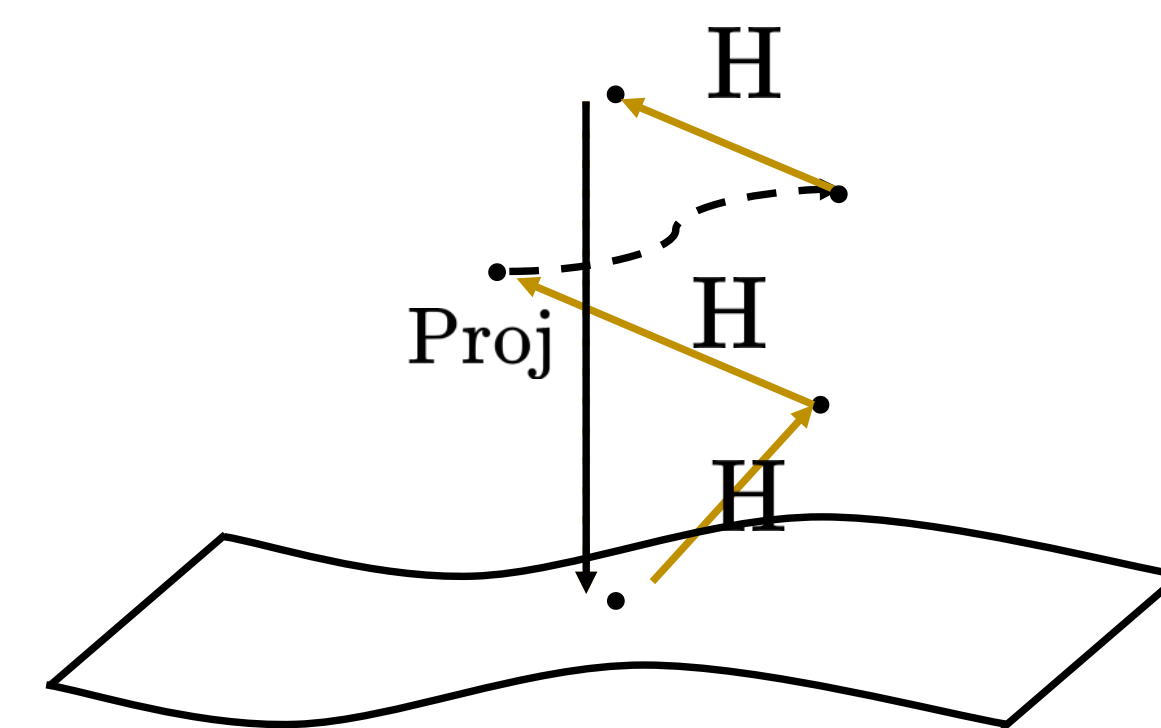
The performance increases with the
depth n used for the target

1.1. Problem: Function approximation

$$q_{\omega} = \text{Proj } H q_{\omega}$$

Q-learning can diverge because $\text{Proj } H$
is not necessarily a contraction

2.1. Main Guarantees



The projected multi-Bellman operator
is a contraction for sufficiently large n

3.1. Bias and Multi-Sampling

$$\max_i \mathbb{E} [Z_i] \neq \mathbb{E} [\max_i Z_i]$$

Multi-step targets can be biased,
so use more m samples of each action

4.1. Conclusion

- Multi Q-learning converges if $n \uparrow$
- The solution is accurate if $m \uparrow$

4.2. Related topics

- Planning and lookahead methods
- Model-based methods

4.3. Future Work

- Non-linear function approximation
- Action pruning methods

Reference: Carvalho, Diogo S., Pedro A. Santos, Francisco S. Melo. "Multi-Bellman operator for convergence of Q-learning with linear function approximation", Transactions on Machine Learning Research, 2025.

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